



Sehat Ledger

Zakat-Preserved Digital Health Camps

An offline-first PWA designed for remote community medical camps, integrating on-device AI diagnosis, localized referral slips, and transparent ledger auditing.

01 / Offline PWA & OCR

Works in zero-network areas.
Scans clinical vitals using
on-device camera.

02 / Multilingual AI Slips

Generates custom Urdu,
Kannada, and English slips
calibrated to low literacy.

03 / Impact Ledger

Translates camp health data
into community care cost
deferred for Zakat audits.

The Problem & Our Innovative Solution

The Challenge

- High Undetected Rates: Diabetes and hypertension go undiagnosed in rural zones.
- Literacy Barriers: Standard medical prescription slips are misunderstood by low-literacy patients.
- Connectivity Blackouts: Rural medical camps frequently lack internet access, rendering traditional cloud software useless.
- Trust & Funding Gaps: Philanthropists lack transparent, audited proof of actual patient outcomes and financial impact.

Our Solution: Sehat Ledger

- Offline-First Engine: Local Node/Browser state allows seamless patient data storage offline.
- Smart Camera Scanning: On-device extraction of glucometer & BP readings using vision AI.
- Tailored LLM Guidance: Translates complex outcomes into warm, supportive messages in native Urdu/Kannada scripts, addressing specific patient barriers.
- Cloud-Sync Gateway: Syncs to Supabase automatically via specialized conflict-free merge RPC scripts once the device re-establishes internet.

Zakat Preservation Ledger

Deferred Care Burden Model (Audit-Ready Financial Math)

Zakat funds exist to preserve community health. Sehat Ledger mathematically proves the direct financial impact of preventative screenings by translating early disease detection into deferred secondary healthcare costs.

Calculation parameters are hardcoded and calibrated strictly to public clinical guidelines (ICMR / MDRF).

Core Financial Formula

1 Dialysis Year = 100 High-Risk Referrals

- Cost of 1 year of dialysis: ~158,880 INR
- Cost per screening: ~15 INR
- Preventative multiplier saved: ~1,588 INR saved for every high-risk case routed to care.

Committee Metrics Verified

- Net Community Burden Deferred (INR)
- Active Referrals Followed-up
- Campaign Zakat Utilization Rate
- Audit trail links every single rupee spent directly to an authenticated clinical patient record.

Why We Win: Tech Stack & Hackathon Edge

The Architecture

- Frontend: Zero-dependency Vanilla HTML5, CSS, and JS (Vibrant Dark Glassmorphism).
- Logic: Pure mathematical clinical rules calibrated to WHO and ICMR guidelines.
- Database & Sync: Supabase REST API + custom conflict-free upsert SQL RPC merger.
- Offline Shell: PWA Service Worker (caches app resources & intercepts offline requests).
- Local Host Server: Lightweight ES module Node.js local test server.

The Hackathon Edge

1. Fully Production-Ready: The app is active, tested, and completely deployed.
2. Solves Real Barriers: Built-in localization ensures patients take action rather than ignoring medical guidance.
3. Financial Auditing: First health-tech app to incorporate Zakat preservation math.

LIVE DEMO LINK:

<https://abf-lab.github.io/Algorism/>